

CAO CLG Notes UNIT 1

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Q: Introduction to Computer Architecture and Organization

Ans:

Computer Architecture and Organization ka use computer systems ko design karne ke liye hota hai. Computer Architecture wo features hoti hain jo user ko visible hote hain, jaise instruction set, addressing modes aur data bits. Ye program ki execution aur system ke behavior ko define karta hai.

Computer Architecture:

- Ye computer system ki design principles aur basic working structure ko describe karta hai.
- Ye hardware ke logical parts par focus karta hai, jaise instruction set aur addressing modes.
- CPU ka kaam kaise hota hai aur memory aur I/O devices ke saath kaise interact karta hai, ye define karta hai.

Computer Organization:

- Ye batata hai ki computer ke hardware parts kaise connected hain aur kaise work karte hain.
- Isme hardware components, data transfer paths, aur control signals ki details include hoti hain.

Q: Difference Between Computer Architecture and Computer Organization

Ans:

Neeche table form me Computer Architecture aur Computer Organization ke beech basic difference simple Hinglish me diye gaye hain (20% short version):

Criteria	Computer Architecture	Computer Organization
Definition	Rules aur design principles jo batate hain computer kaise work karta hai. System ki logical structure aur behavior define karta hai.	Batata hai hardware components kaise connect hote hain aur kaise work karte hain. Practical implementation.
Focus	Instruction set, data types, addressing modes, system design.	Implementation details jaise data paths, control signals, physical components.
Concerned With	High-level design like ISA and performance optimization.	Low-level hardware details like circuits, ALU, microarchitecture.
Design Aspects	Instruction set, processor capabilities, memory hierarchy.	Control unit, bus system, I/O handling, internal data transfer.
User Perspective	Software developers ko visible. Programmer interface define karta hai.	Mostly hardware designers ke liye useful, efficient hardware usage par focus.
Example Components	Instruction set, data format, pipelining, parallelism.	Registers, cache design, control signals, timing signals.
Design Level	Abstract aur logical high-level design.	Physical implementation aur low-level technical design.
Examples	RISC vs CISC, Harvard vs Von Neumann architecture.	Intel/AMD microarchitecture, ALU design, control unit organization.

Agar next topic ready ho, bhej do 😊✨

Q: CPU, Memory, I/O Devices, Buses & Types of CPU Buses with Working Cycle

Ans:

CPU (Central Processing Unit):

Computer ka brain hota hai. Isme CU (Control Unit) aur ALU (Arithmetic Logic Unit) hote hain jo instructions ko decode aur execute karte hain aur calculations perform karte hain.

Memory:

Ek memory unit jo program ke instructions aur data ko store karti hai.

I/O Devices:

Input aur Output devices user aur computer ke beech communication provide karte hain.

Buses:

Communication pathways hote hain jo CPU, memory aur I/O devices ke beech data transfer karte hain.

Types of CPU Buses

Address Bus:

One-way bus hota hai jo CPU se memory ya I/O tak address carry karta hai, batata hai data kahan se lana ya bhejna hai.

Data Bus:

Two-way bus jo actual data transfer karta hai CPU aur memory/I/O ke beech.

Control Bus:

Control signals carry karta hai jo system ki operations ko manage aur synchronize karta hai, jaise read ya write.

How It Works (Working Cycle)

1. **Instruction Fetch:** CPU memory se instruction fetch karta hai.
2. **Instruction Decode:** Control Unit instruction ko decode karti hai.
3. **Data Fetch:** Agar data chahiye, to CPU memory se fetch karta hai.
4. **Execution:** ALU operation perform karta hai.
5. **Data Storage:** Result memory me store hota hai.
6. **Sequential Cycle:** Phir next instruction ke liye cycle repeat hota hai.

Done! Next content bhejo 🤖✨

Q: Throughput, Latency, MIPS, FLOPS

Ans:

1. Throughput

Definition:

Throughput matlab system ek time period me kitna work complete karta hai ya kitna data process karta hai.

Characteristics:

- Tasks per second, TPS, Bps jaise units me measure hota hai.
- High throughput matlab system zyada load handle kar sakta hai.
- Ye individual task time ko count nahi karta.

Example:

- Web server 1000 request per second handle karta hai to throughput = 1000 TPS.
- Network 500 Mbps transfer karta hai to throughput 500 Mbps.

Use Case:

Databases, web servers, networks aur processing systems ki performance measure karne ke liye.

2. Latency

Definition:

Latency matlab kisi request ka response aane me lagne wala time.

Characteristics:

- Milliseconds ya microseconds me measure hota hai.

- Low latency matlab fast response.
- Network delays, processing time, I/O par depend karta hai.

Example:

- Ping response ka time latency hota hai.
- Database query ka response time bhi latency hota hai.

Use Case:

Real-time systems, gaming, streaming, trading jaha quick response important hota hai.

3. MIPS (Million Instructions Per Second)

Definition:

Processor ek second me kitne million machine instructions run karta hai.

Characteristics:

- Millions instructions per second me measure hota hai.
- Processor ki raw speed batata hai.
- Architecture compare karne ke liye hamesha accurate nahi.

Example:

- CPU 3 MIPS matlab 3 million instructions per second.
- Microcontroller 2.5 MIPS matlab 2.5 million instructions per second.

Use Case:

Benchmarking, old processors, microcontrollers, embedded systems.

Limitations:

- Complex instructions time differ karte hain.
 - RISC vs CISC architecture compare karne me useful nahi.
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4. FLOPS (Floating-Point Operations Per Second)

Definition:

Floating-point mathematical calculations perform karne ki speed measure karta hai.

Characteristics:

- GFLOPS, TFLOPS, PFLOPS units me measure hota hai.
- Supercomputers, GPUs, scientific aur AI tasks ke liye important.
- Complex calculations ke performance ko show karta hai.

Example:

- 1 TFLOPS processor 1 trillion floating-point operations per second perform karta hai.
- NVIDIA GPU 15 TFLOPS deep learning ke liye powerful hota hai.

Use Case:

Supercomputers, simulations, machine learning, graphics processing.

Limitations:

- Sirf floating-point operations evaluate karta hai.
- Performance hardware aur parallel processing par depend karta hai.

Next topic bhejo 😊✨

Q: Comparison Table - Throughput, Latency, MIPS, FLOPS

Ans:

Aspect	Throughput	Latency	MIPS	FLOPS
Definition	Kaam ya data processing rate per time unit	Task start se complete hone tak ka delay	Machine instructions per second execution speed	Floating-point operations execution speed
Unit	Tasks/sec, Bytes/sec, Requests/sec	ms, μ s, sec	Million Instructions Per Second	Floating-Point Ops Per Second
Focus	System capacity aur volume	Response time	Instruction execution rate	Floating-point calculation performance
Use Case	Databases, networks, web servers	Games, streaming, real-time apps	Embedded systems, CPUs	Supercomputers, GPUs, scientific work
Limitation	Individual task time show nahi karta	System capacity show nahi karta	Different architectures comparison inaccurate	Sirf floating-point performance measure karta

Examples & Use Cases

1. Throughput (Database):

Database 5000 TPS handle kare to high throughput show hota hai.

2. Latency (Gaming):

Online gaming me 20 ms se kam latency ideal hoti hai, warna lag feel hota hai.

3. MIPS (Microcontroller):

50 MIPS microcontroller 50 million instructions per second execute karta hai, sensor processing ke liye useful.

4. FLOPS (Scientific Computing):

1 PFLOPS supercomputer complex simulations jaise weather prediction ke liye perfect hai.

Next bhejo bhai 😊✨

Q: SISD, SIMD, MISD, MIMD Architectures

Ans:

1. SISD (Single Instruction Single Data)

Description:

Ek single processor ek time me ek instruction ko ek data stream par execute karta hai. Matlab ek CPU sequential instructions run karta hai aur ek hi data set par kaam karta hai.

Example:

Normal desktop ya laptop CPU jo ek program ko sequentially execute karta hai.

2. SIMD (Single Instruction Multiple Data)

Description:

Ek control unit same instruction ko multiple processors ko send karti hai aur ye processors different data par parallel processing karte hain.

Example:

GPU, vector processors jahan ek hi operation (jaise pixel addition) bohot sare data par ek saath hota hai.

3. MISD (Multiple Instruction Single Data)

Description:

Multiple processors different instructions run karte hain, lekin sab ek hi data stream par kaam karte hain.

Example:

Commercial systems me rare hota hai. Systolic arrays signal processing jaisi tasks me use hote hain.

4. MIMD (Multiple Instruction Multiple Data)

Description:

Multiple processors independent kaam karte hain, har ek apna instruction aur apna data handle karta hai. True multitasking allow karta hai.

Example:

Modern multi-core systems, smartphones, laptops.

Next topic bhejo! 😊✨

Q: Comparison Table - SISD, SIMD, MIMD, MISD

Ans:

Type	Description	Example	Usage
SISD	Ek processor ek instruction ko ek data stream par execute karta hai.	Single-core CPUs	Normal desktop aur laptop systems
SIMD	Same instruction multiple data streams par ek saath run hota hai.	GPUs	Image processing, scientific computing
MIMD	Multiple processors different instructions aur different data par kaam karte hain.	Multicore processors like Intel i7, AMD Ryzen	Parallel processing, supercomputers, multicore systems
MISD	Multiple processors different instructions ko same data par execute karte hain.	Fault-tolerant systems (rare)	Critical systems, error checking

Next topic ready ho to bhej do bhai 😊✨

Q: Machine Language and Assembly Language

Ans:

1. Machine Language

Definition:

Machine language sabse low-level programming language hoti hai jo sirf binary code (0s aur 1s) se bani hoti hai. CPU ise directly execute karta hai bina kisi translation ke.

Characteristics:

- Binary format me instructions (0s aur 1s).
- Hardware-specific (e.g., Intel x86, ARM).
- Likhna aur samajhna bohot difficult aur error-prone.

Advantages & Disadvantages

Advantages	Disadvantages
Direct CPU execution	Likhna complex aur error-prone
Fast and efficient	Hardware-specific, portable nahi
Compiler ya assembler ki need nahi	Debugging mushkil

2. Assembly Language

Definition:

Assembly language low-level language hai jo machine code ke jagah mnemonics (MOV, ADD, SUB, JMP) use karti hai. Ye machine language se readable hoti hai aur assembler use karke machine code me convert hoti hai.

Characteristics:

- Mnemonics use karti hai, binary se easy hota hai.
- Labels aur symbolic names allow karti hai.
- CPU architecture specific.
- Execution ke liye assembler chahiye.

Advantages & Disadvantages

Advantages	Disadvantages
Machine language se easy aur readable	High-level languages ke comparison me still difficult
Fast, hardware control allow karti hai	Portable nahi
Performance-critical code ke liye useful	Architecture knowledge essential aur error-prone

Next topic bhejo 😊✨

Q: Machine Language vs Assembly Language

Ans:

Aspect	Machine Language	Assembly Language
Representation	Binary code (0s, 1s)	Mnemonics like MOV, ADD, SUB
Readability	Human-readable nahi	More readable than machine language
Translation	Direct CPU execution	Assembler ke through machine code me convert
Ease of Use	Bohot difficult aur error-prone	Easy comparatively, but still complex
Portability	CPU-specific	CPU-specific
Example	10101000 00000101	MOV AX, 5

Use Cases

Machine Language:

- Direct hardware control jaha speed critical hai.
- Bootloaders, firmware, microcode.

Assembly Language:

- Embedded systems, device drivers, real-time systems.
- Performance-critical code like microcontroller programming, BIOS development.

Next topic bhej do 😊✨

Q: What is Floating-Point Arithmetic?

Ans:

Floating-point arithmetic real numbers (fractional values) ko represent aur calculate karne ka tarika hai. Ye scientific notation jaise format use karta hai jisme number ko is form me likha jata hai:

$$\text{Number} = \text{Sign} \times \text{Mantissa} \times 2^{\text{Exponent}}$$

Components

1. Sign Bit:

Batata hai number positive hai ya negative.

- 0 = Positive
- 1 = Negative

2. Mantissa:

Number ke significant digits show karta hai. Precision decide karta hai.

3. Exponent:

Power of 2 represent karta hai jisse large ya small values ko represent karna possible hota hai.

Example

Decimal number: **-13.25**

1. Binary conversion: **-1101.01**
2. Scientific form: **-1.10101 × 2³**
3. Components:
 - **Sign bit:** 1
 - **Mantissa:** 1.10101
 - **Exponent:** 3

Next topic do 😊✨

Q: IEEE 754 Standard for Floating-Point Arithmetic

Ans:

IEEE 754 ek standard hai jo floating-point numbers ko represent aur process karne ke rules define karta hai. Ye ensure karta hai ki alag-alag systems par results same aur accurate aayein.

Key Features

- Single-precision (32-bit) aur Double-precision (64-bit) formats provide karta hai.
- Special values define karta hai: **Infinity, NaN, Zero, Denormal numbers.**
- Rounding, overflow, underflow aur errors ke rules set karta hai.

1. Why Use Signed Number Representation?

- Binary sirf 0 aur 1 se bana hota hai, isliye normally sirf positive numbers represent ho sakte hain.
- Negative numbers show karne ke liye special method chahiye hota hai.
- Signed number representation binary patterns use karke negative values ko represent karna possible banata hai.

Next topic bhejo 😊✨